

Linux task 03:-

Part A: Process Monitoring

```
Session Actions Edit View Help
[~] (sneha@kali) ~
$ ps
  PID TTY          TIME CMD
 298 pts/1    00:00:00 ps
 303 pts/1    00:00:00 ps

[~] (sneha@kali) ~
$ ps aux
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0  0.7 24628 15920 ?        Ss   06:54   0:01 /sbin/init splash
root         2   0.0  0.0      0   0 0?        Ss   06:54   0:00 [kthreadd]
root         3   0.0  0.0      0   0 0?        Ss   06:54   0:00 [pool_workqueue_release]
root         4   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-rcu_gp]
root         5   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-sync_wq]
root         6   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-free_rcu_reclaim]
root         7   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-slab_flushwq]
root         8   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-netns]
root         9   0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/8-cgroup_free]
root        10   1.3  0.0      0   0 0?        Ic   06:54   0:01 [kworker/0:hw-kslockd]
root        11   0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/0:1-events]
root        12   0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/u12:8-ipv6_addrconf]
root        13   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-m_percpu_wq]
root        14   0.0  0.0      0   0 0?        Ss   06:54   0:00 [ksftirqd/0]
root        15   0.1  0.0      0   0 0?        Ss   06:54   0:00 [rcu_preempt]
root        16   0.0  0.0      0   0 0?        Ss   06:54   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        17   0.1  0.0      0   0 0?        Ss   06:54   0:00 [rcu_exp_gp_kthread_worker]
root        18   0.0  0.0      0   0 0?        Ss   06:54   0:00 [migration/0]
root        19   0.0  0.0      0   0 0?        Ss   06:54   0:00 [idle_inject/0]
root        20   0.0  0.0      0   0 0?        Ss   06:54   0:00 [cpuhp/0]
root        21   0.0  0.0      0   0 0?        Ss   06:54   0:00 [cpuhp/1]
root        22   0.0  0.0      0   0 0?        Ss   06:54   0:00 [idle_inject/1]
root        23   0.1  0.0      0   0 0?        Ss   06:54   0:00 [migration/1]
root        24   0.1  0.0      0   0 0?        Ss   06:54   0:00 [ksftirqd/1]
root        25   0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/1:8-ata_sff]
root        26   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/2:0-hw-kslockd]
root        27   0.0  0.0      0   0 0?        Ss   06:54   0:00 [cpuhp/2]
root        28   0.0  0.0      0   0 0?        Ss   06:54   0:00 [idle_inject/2]
root        29   0.1  0.0      0   0 0?        Ss   06:54   0:00 [migration/2]
root        30   0.0  0.0      0   0 0?        Ss   06:54   0:00 [ksftirqd/2]
root        31   0.1  0.0      0   0 0?        I   06:54   0:00 [kworker/2:8-ata_sff]
root        32   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/2:0-hw-kslockd]
root        33   1.4  0.0      0   0 0?        I   06:54   0:01 [kworker/u1:8-events_unbound]
root        34   0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/u4:0-ata-rsv-conversion]
root        35   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/u5:0-flush-0]
root        37   0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/u5:1-kvfree_rcu_reclaim]
root        38   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/2]
root        39   0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-lsm_frag_wq]
root        40   0.0  0.0      0   0 0?        Ss   06:54   0:00 [rcu_tasks_kthread]
root        41   0.0  0.0      0   0 0?        I   06:54   0:00 [rcu_tasks_rude_kthread]
root        42   0.0  0.0      0   0 0?        I   06:54   0:00 [rcu_tasks_trace_kthread]
```

```
Session Actions Edit View Help
[~] (sneha@kali) ~
root      2970  0.0  0.0      0   0 0?        Ic   06:55   0:00 [kworker/12H]
sneha     3092  0.0  0.2  9540 4460 pts/1    Re   06:55   0:00 ps aux

[~] (sneha@kali) ~
$ top
top - 06:56:38 up 2 min, 1 user, load average: 1.59, 1.00, 0.42
Tasks: 218 total, 1 running, 217 sleeping, 0 stopped, 0 zombie
MiB Mem:  0.0 ul, 1.7 fr, 0.0 ca, 98.5 us;  0.0 vs, 0.0 hi, 0.0 si, 0.0 st
MiB Mem:  1971.5 total,  166.6 free,  1326.3 used,  715.1 buff/cache
MiB Swap: 1181.0 total,  731.2 free,  450.7 used,  645.2 avail Mem

  PID  PPID  %CPU   %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
 780 root      0   0.0  0.0 541388 202384 102932 S   2.0 10.0  0:18.27 Xorg
1056 sneha    20   0.0  0.0 315380 3800 2932 S   0.7 0.1  0:00.75 VBoxClient
1240 sneha    20   0.0  0.0 307956 3616 19720 S   0.7 1.0  0:02.10 xorgxrdp.0
 63 root      0   0.0  0.0      0   0 0?        Ss   06:54   0:00 systemd-journal
 616 root      0   0.0  0.0      0   0 0?        Ss   06:54   0:00 systemd-logd
1125 root      0   0.0  0.0      0   0 0?        Ss   06:54   0:00 [kworker/2:3-events]
1128 sneha    20   0.0  0.0 208032 95020 82352 S   0.2 0.7  0:02.42 ffmpeg
1578 sneha    39 19 272288 26216 19892 S   0.3 1.4  0:00.33 tmux
1818 sneha    20   0.0  0.0 11336 508308 35314 S   0.3 22.3  0:44.47 firefox-esr
2038 sneha    20   0.0  0.0 245284 119124 99096 S   0.1 5.9  0:01.17 Privileged Cont
2327 sneha    20   0.0  0.0 2487732 116536 99696 S   0.3 5.0  0:01.53 file:/// Content
2617 sneha    20   0.0  0.0 74884 72924 55496 S   0.1 1.6  0:02.35 sterminal
3403 sneha    20   0.0  0.0 10464 6080 3908 R   0.3 0.3  0:00.07 top
  1 root      0   0.0  0.0 24628 15920 11580 S   0.0 0.0  0:01.39 systemd
  2 root      0   0.0  0.0      0   0 0?        Ss   06:54   0:00 [kthreadd]
  3 root      0   0.0  0.0      0   0 0?        Ss   06:54   0:00 [pool_workqueue_release]
  4 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-rcu_gp]
  5 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-sync_wq]
  6 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-free_rcu_reclaim]
  7 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-slab_flushwq]
  8 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-netns]
  9 root      0  -20  0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/8-cgroup_free]
 10 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:01.29 [kworker/0:hw-kslockd]
 11 root      0  -20  0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/0:1-events]
 12 root      0  -20  0.0  0.0      0   0 0?        I   06:54   0:00 [kworker/u12:8-ipv6_addrconf]
 13 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00 [kworker/8-m_percpu_wq]
 14 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.82 [ksftirqd/0]
 15 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.22 [rcu_preempt]
 16 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.00 [rcu_exp_par_gp_kthread_worker/0]
 17 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.16 [rcu_exp_gp_kthread_worker]
 18 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.82 [migration/0]
 19 root     -51  0.0  0.0      0   0 0?        Ss   06:54   0:00.00 [idle_inject/0]
 20 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.00 [cpuhp/0]
 21 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.00 [idle_inject/1]
 22 root     -51  0.0  0.0      0   0 0?        Ss   06:54   0:00.00 [cpuhp/1]
 23 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.19 [migration/1]
 24 root      0  -20  0.0  0.0      0   0 0?        Ss   06:54   0:00.22 [ksftirqd/1]
 25 root      0  -20  0.0  0.0      0   0 0?        Ic   06:54   0:00.00 [kworker/1:8-ata_sff]
```

```

sneh@kali:~$
Session Actions Edit View Help

4.7% Tasks: 356, 593 ty, 106 other, 3 running
4.7% Load average: 0.46 0.76 0.41
4.5% Uptime: 00:05:29
2007.4303
322M/1.64G

Main View
PID USER %MEM
180 root 0
1061 sneh 0
1069 sneh 0
630 root 0
1128 sneh 0
1178 sneh 0
1184 sneh 0
1192 sneh 0
2017 sneh 0
7022 root 0
1 root 0
333 root 0
373 root 0
375 root 0
400 root 0
486 root 0
495 polkit 0
548 root 0
529 root 0
530 root 0
553 root 0
572 root 0
580 polkit 0
581 polkit 0
586 polkit 0
616 root 0
622 root 0
624 root 0
625 root 0
627 root 0
628 root 0
629 root 0
632 root 0
635 root 0
636 root 0
640 root 0
643 root 0
642 root 0
645 root 0

TIME: Command
0.0 0.00.00 /usr/bin/sudoedit -drapdrop
0.0 0.00.00 /usr/bin/dbus-daemon --system --address=systemd; --nofork --nopidfile --systemd-activation --syslog-only
0.0 0.00.03 /usr/sbin/ModemManager
0.0 0.00.45 xfsmk
0.0 0.00.21 xfsmk
0.0 0.00.03 Thunder -daemon
0.0 0.00.03 /usr/libexec/updrrd
0.0 0.00.05 /usr/bin/terminal
0.0 0.00.10 ls
0.0 0.00.13 /sbin/init splash
0.0 0.00.30 /usr/lib/systemd/systemd-journald
0.0 0.00.30 /usr/lib/systemd/systemd-udevd
0.0 0.00.03 /usr/lib/systemd/systemd-userdbd
0.0 0.00.02 /usr/sbin/hwlogd -foreground -verbose=1
0.0 0.00.11 /usr/libexec/accounts-daemon
0.0 0.00.34 /usr/lib/polkit-1/polkitd --no-debug --log-lev=notice
0.0 0.00.24 /usr/lib/systemd/systemd-logind
0.0 0.00.00 /usr/libexec/accounts-daemon
0.0 0.00.00 /usr/libexec/accounts-daemon
0.0 0.00.03 /usr/libexec/accounts-daemon
0.0 0.00.27 /usr/sbin/NetworkManager --no-daemon
0.0 0.00.02 /usr/lib/polkit-1/polkitd --no-debug --log-lev=notice
0.0 0.00.00 /usr/lib/polkit-1/polkitd --no-debug --log-lev=notice
0.0 0.00.11 /usr/lib/polkit-1/polkitd --no-debug --log-lev=notice
0.0 0.00.25 /usr/bin/VBoxSvcBMCClient
0.0 0.00.00 /usr/bin/VBoxSvcBMCClient
0.0 0.00.21 /usr/bin/VBoxSvcBMCClient
0.0 0.00.00 /usr/sbin/WiMoService
0.0 0.00.00 /usr/sbin/WiMoService
0.0 0.00.00 /usr/sbin/WiMoService
0.0 0.00.00 /usr/sbin/WiMoService
0.0 0.00.00 /usr/sbin/WiMoService
0.0 0.00.15 /usr/sbin/ModemManager
0.0 0.00.01 /usr/sbin/WiMoService
0.0 0.00.01 /usr/sbin/WiMoService
0.0 0.00.00 /usr/bin/VBoxSvcBMCClient
0.0 0.00.00 /usr/bin/VBoxSvcBMCClient
0.0 0.00.00 /usr/sbin/WiMoService
0.0 0.00.00 /usr/sbin/NetworkManager --no-daemon

File Help Setup Search Filter View SortBy Nice Police Pull Filter

```

1. What is a Process?

A process is a program that is currently running on the computer. Every application or command that is executed creates a process.

2. What is a PID?

PID stands for **Process ID**. It is a unique number assigned by the operating system to identify each running process.

3. Which process is consuming the most CPU?

The process consuming the most CPU is **firefox-esr** with **0.3% CPU usage**.

4. Which process is consuming the most Memory?

The process with the highest value is consuming the most memory. **firefox-esr** with **22.3% memory usage**. Is the process consuming the most memory.

Part B: Process Management

```
Session Actions Edit View Help
(sneha@kali)-[~]
└─$ sleep 300 &
[1] 20605

(sneha@kali)-[~]
└─$ ps aux |grep sleep
sneha 19077 0.0 0.1 5616 2100 pts/2 SN 07:20 0:00 sleep 300
sneha 19883 0.0 0.1 5616 2172 pts/2 SN 07:22 0:00 sleep 300
sneha 20605 0.0 0.1 5616 2236 pts/3 SN 07:23 0:00 sleep 300
sneha 20696 0.0 0.1 6544 2384 pts/3 S+ 07:23 0:00 grep --color=auto sleep

(sneha@kali)-[~]
└─$ kill 19077

(sneha@kali)-[~]
└─$ ps aux |grep sleep
sneha 19883 0.0 0.1 5616 2172 pts/2 SN 07:22 0:00 sleep 300
sneha 20605 0.0 0.1 5616 2236 pts/3 SN 07:23 0:00 sleep 300
sneha 20892 0.0 0.1 6544 2440 pts/3 S+ 07:24 0:00 grep --color=auto sleep

(sneha@kali)-[~]
└─$ kill -9 19883

(sneha@kali)-[~]
└─$ ps aux |grep sleep
sneha 20605 0.0 0.1 5616 2236 pts/3 SN 07:23 0:00 sleep 300
sneha 21127 0.0 0.1 6544 2388 pts/3 S+ 07:24 0:00 grep --color=auto sleep

(sneha@kali)-[~]
└─$
```

PID Found:

3456

Command Used:

ps aux | grep sleep

kill 3456

Result:

The sleep 300 process was successfully found using the ps aux | grep sleep command. The process was terminated using the kill command and stopped running successfully.

Part C: System Monitoring

```

sneha@kali: ~
$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi          1.4Gi          124Mi          80Mi          651Mi          522Mi
Swap:          1.1Gi          476Mi          624Mi                   0

sneha@kali: ~
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            883M    0   883M   0% /dev
tmpfs           198M 1004K   197M   1% /run
/dev/sda1       19G   16G   2.3G  88% /
tmpfs           986M   4.0K   986M   1% /dev/shm
none            1.0M    0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           986M   20K   986M   1% /tmp
none            1.0M    0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  112K   198M   1% /run/user/1000

sneha@kali: ~
$ uptime
07:32:50 up 38 min,  1 user,  load average: 0.88, 0.54, 0.58

sneha@kali: ~
$ uname -a
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux

sneha@kali: ~
$
```

Records are as follows:

- Total RAM :-1.9Gi
- Available RAM :-1.4Gi
- Disk Usage :-88%
- System Uptime :-38 min
- Kernel Version :-6.18.12+kali-amd64

System Summary Report

Commands Used

free -h
df -h
uptime
uname -a

Memory Information

- **Total RAM:** 1.9 GiB
- **Available RAM:** 1.4 GiB

Disk Information

- **Disk Usage:** 88%

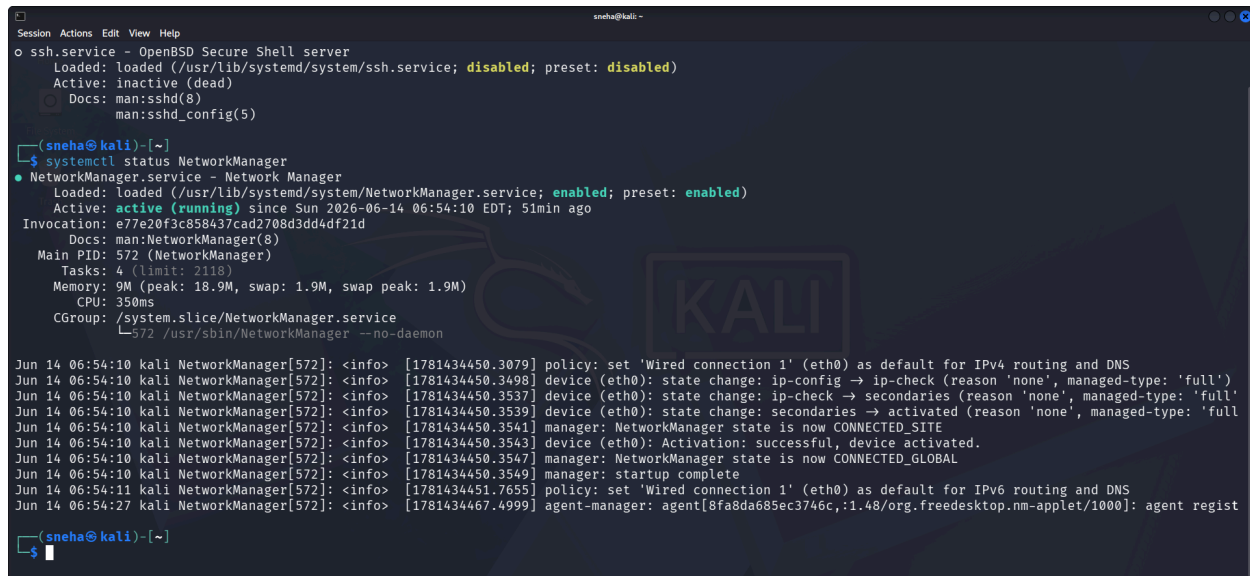
System Information

- **System Uptime:** 38 minutes
- **Kernel Version:** 6.18.12+kali-amd64

Summary

The system information was collected using Linux monitoring commands. The system has **1.9 GiB** of total RAM, with **1.4 GiB** currently available. The disk usage is **88%**. The system has been running for **38 minutes**. The installed Linux kernel version is **6.18.12+kali-amd64**.

Part D: Service Monitoring

A terminal window on a Kali Linux system. The user runs 'systemctl status ssh', showing it's loaded but inactive. Then they run 'systemctl status NetworkManager', showing it's loaded and active (running). Below this, a series of log messages from NetworkManager are displayed, showing the process of activating the network interface eth0, including setting policies for IPv4 and IPv6, and activating the device.

```
Session Actions Edit View Help
o ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
  Active: inactive (dead)
  Docs: man:sshd(8)
       man:sshd_config(5)

(sneha@kali)~$ systemctl status ssh
● NetworkManager.service - Network Manager
  Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: enabled)
  Active: active (running) since Sun 2026-06-14 06:54:10 EDT; 51min ago
  Invocation: e77e20f3c858437cad2708d3dd4df21d
  Docs: man:NetworkManager(8)
  Main PID: 572 (NetworkManager)
  Tasks: 4 (limit: 2118)
  Memory: 9M (peak: 18.9M, swap: 1.9M, swap peak: 1.9M)
  CPU: 350ms
  CGroup: /system.slice/NetworkManager.service
          └─572 /usr/sbin/NetworkManager --no-daemon

Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3079] policy: set 'Wired connection 1' (eth0) as default for IPv4 routing and DNS
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3498] device (eth0): state change: ip-config → ip-check (reason 'none', managed-type: 'full')
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3537] device (eth0): state change: ip-check → secondaries (reason 'none', managed-type: 'full')
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3539] device (eth0): state change: secondaries → activated (reason 'none', managed-type: 'full')
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3541] manager: NetworkManager state is now CONNECTED_SITE
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3543] device (eth0): Activation: successful, device activated.
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3547] manager: NetworkManager state is now CONNECTED_GLOBAL
Jun 14 06:54:10 kali NetworkManager[572]: <info> [1781434450.3549] manager: startup complete
Jun 14 06:54:11 kali NetworkManager[572]: <info> [1781434451.7655] policy: set 'Wired connection 1' (eth0) as default for IPv6 routing and DNS
Jun 14 06:54:27 kali NetworkManager[572]: <info> [1781434467.4999] agent-manager: agent[8fa8da685ec3746c,:1.48/org.freedesktop.nm-applet/1000]: agent regist
```

Commands Used

```
systemctl status ssh
systemctl status NetworkManager
```

1.What is a Service?

A service is a program that runs in the background and performs specific tasks for the operating system or users. Services usually start automatically when the system boots.

2.Why are Services Important?

Services are important because they provide essential functions such as networking, remote access, printing, and system management. Many system operations depend on services running correctly.

3.How Can a Stopped Service Affect a System?

If an important service stops running, related system functions may stop working. For example, if the NetworkManager service is stopped, network connectivity may be affected. Similarly, if the SSH service is stopped, remote access to the system may not be available.

Part F: Security Monitoring Challenge

1. netstat

Purpose

`netstat` is used to display network connections, routing tables, interface statistics, and listening ports.

Example Output

```
tcp  0  0 0.0.0.0:22  0.0.0.0:*  LISTEN
```

Security Use Case

Security professionals use `netstat` to identify open ports and active network connections that may indicate unauthorized access or suspicious activity.

2. ss

Purpose

`ss` is used to display socket statistics and active network connections. It is a faster replacement for `netstat`.

Example Output

```
tcp  LISTEN  0  128  0.0.0.0:22  0.0.0.0:*
```

Security Use Case

It helps security analysts monitor open ports, active connections, and network services running on a system.

3. who

Purpose

`who` displays information about users currently logged into the system.

Example Output

```
sneha pts/0 2026-06-14 08:30
```

Security Use Case

Administrators can verify who is currently logged in and detect unauthorized user sessions.

4. w

Purpose

`w` shows information about logged-in users and their current activities.

Example Output

```
USER  TTY  LOGIN@  IDLE   JCPU   PCPU   WHAT
sneha pts/0 08:30   1:00   0.05s  0.02s  bash
```

Security Use Case

It helps monitor user activity and identify suspicious or unexpected processes being run by users.

5. last

Purpose

`last` displays the login history of users on the system.

Example Output

```
sneha pts/0 192.168.1.10 Sun Jun 14 08:30
```

Security Use Case

Security teams use `last` to review login history, investigate unauthorized access attempts, and perform forensic analysis after security incidents.

Part G: Mini SOC Activity

1.How would you identify resource-heavy processes?

I will use commands like `top`, `htop`, and `ps aux` to see all running processes. These commands show how much CPU and memory each process is using. If a process is using too much CPU or memory, it may make the system slow.

2.How would you determine whether a process is suspicious?

I will check the process name and its usage. If a process has a strange name or uses too many resources without any reason, it may be suspicious. I will also check if it belongs to a known program.

3.What information would you collect before terminating a process?

Before stopping a process, I will collect:

- Process Name
- PID (Process ID)
- CPU Usage
- Memory Usage
- User Name

This information helps me understand the process and avoid stopping an important system process.

4.Conclusion

If a system is running slowly, I will first check the running processes. Then I will find the process using too many resources and check whether it is safe or suspicious before stopping it.